Eq. (6.4.6) then becomes:

$$\left[\frac{\delta^r S[\epsilon]}{\delta\epsilon_{a_1}(x_1)\cdots\delta\epsilon_{a_r}(x_r)}\right]_{\epsilon=0} = \sum_{N=0}^{\infty}\frac{(-i)^{N+r}}{N!}\sum_{N_0N_1\cdots N_r}\frac{N!\,\delta_{N,N_0+N_1+\cdots+N_r}}{N_0!N_1!\cdots N_r!}$$
$$\times\int_{x_1^0}^{\infty}d\tau_{01}\cdots d\tau_{0N_0}\int_{x_2^0}^{x_1^0}d\tau_{11}\cdots d\tau_{1N_1}\cdots\int_{-\infty}^{x_r^0}d\tau_{r1}\cdots d\tau_{rN_r}$$
$$\times\Big(\Phi_\beta,\ T\Big\{V(\tau_{01})\cdots V(\tau_{0N_0})\Big\}\,o_{a_1}(x_1)T\Big\{V(\tau_{11})\cdots V(\tau_{1N_1})\Big\}\,o_{a_2}(x_2)\cdots$$
$$\times\cdots o_{a_r}(x_r)\ T\Big\{V(\tau_{r1})\cdots V(\tau_{rN_r})\Big\}\Phi_\alpha\Big)\,.$$

The factor $N!/N_0!N_1!\cdots N_r!$ is the number of ways of sorting N τs into $r+1$ subsets, each containing N_0, $N_1,\cdots N_r$ of these τs. Instead of summing over $N_0, N_1,\cdots,N_r$, subject to the condition $N_0+N_1+\cdots+N_r = N$, and then summing over N, we can just sum independently over N_0, $N_1,\cdots,N_r$, setting N where it appears in $(-i)^N$ equal to $N_0+N_1+\cdots+N_r$. This gives

$$\left[\frac{\delta^r S[\epsilon]}{\delta\epsilon_{a_1}(x_1^0)\cdots\delta\epsilon_{a_r}(x_r^0)}\right]_{\epsilon=0} = (-i)^r\Big(\Phi_\beta, U(\infty,x_1^0)\,o_{a_1}(x_1)$$
$$\times\ U(x_1^0,x_2^0)\,o_{a_2}(x_2)\,U(x_2^0,x_3^0)\cdots o_{a_r}(x_r)\,U(x_r^0,-\infty)\Phi_\alpha\Big)\,, \tag{6.4.7}$$

where

$$U(t',t) = \sum_{N=0}^{\infty}\frac{(-i)^N}{N!}\int_t^{t'}d\tau_1\cdots d\tau_N\ T\Big\{V(\tau_1)\cdots V(\tau_N)\Big\}\,. \tag{6.4.8}$$

The operator $U(t',t)$ satisfies the differential equation

$$\frac{d}{dt'}U(t',t) = -iV(t')U(t',t) \tag{6.4.9}$$

with the obvious initial condition

$$U(t,t) = 1\,. \tag{6.4.10}$$

This has the solution

$$U(t',t) = \exp(iH_0t')\exp(-iH(t'-t))\exp(-iH_0t) = \Omega^{-1}(t')\Omega(t) \tag{6.4.11}$$

with Ω given by Eq. (6.4.5). Inserting Eq. (6.4.11) in Eq. (6.4.7) and using